



EZITECH ENGINEERING FRAMEWORK • CASE STUDY AI-005

AI-Powered Internship Performance Prediction & Risk Analytics Platform

Predicting dropout risk and performance trends before problems happen — so mentors can act early, not late.



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Team Size: 2 AI Engineers

Duration: 4 Weeks

The Problem

Manual performance monitoring doesn't scale with internship growth



Late Detection

Struggling interns often aren't noticed until it's too late



High Dropout

Manual monitoring can't keep pace with growing batch sizes



Delayed Intervention

Mentors respond reactively instead of catching risk early



No Predictive Analytics

Nothing forecasts risk before it becomes visible on its own



No Personalization

Interns get generic guidance instead of tailored insight



Manual Reporting

Management reporting eats time better spent mentoring



Ezitech manages hundreds of interns **across Laravel, MERN, AI, Flutter, UI/UX, and DevOps** — manual monitoring simply doesn't scale.

Project Objective

Design an AI-driven platform that monitors intern activity, predicts future performance, detects risk, and recommends personalized interventions — before issues occur.



1. Monitor

Continuously track attendance, tasks, GitHub activity, code reviews, and communication



2. Predict

Forecast dropout risk, performance trend, and success probability with explainable ML



3. Recommend

Turn every prediction into a concrete, personalized intervention for mentors

What We Built

An end-to-end predictive analytics platform, not a proof-of-concept



8 Prediction Types

Dropout risk, performance trend, success probability, skill growth & more



Explainable AI

Every prediction ships with SHAP-based reasoning, not just a score



Recommendation Engine

Rule-based actions wired directly to live model outputs



3 Role-Based Dashboards

Purpose-built views for Admin, Mentor, and Student



Documented REST API

FastAPI backend with full OpenAPI / Swagger reference



Full Monitoring Stack

MLflow tracking + Prometheus & Grafana observability

System Architecture



Cross-cutting infrastructure: PostgreSQL + Redis for storage · MLflow for experiment tracking · Prometheus + Grafana for live monitoring · Docker Compose for orchestration

Data & Feature Engineering

Six live signal sources feed a single, leakage-safe feature pipeline

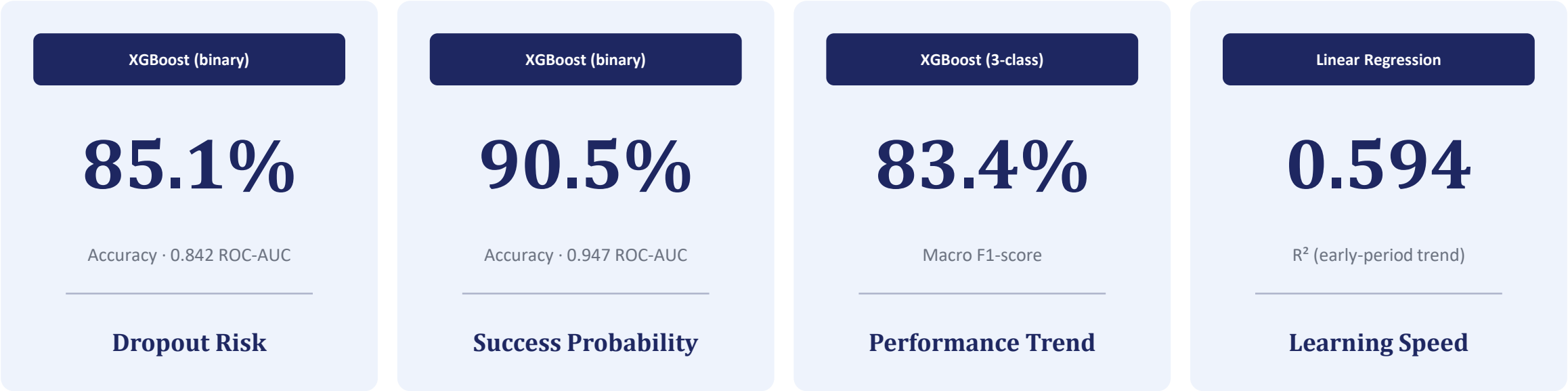


Why the pipeline matters

- Shared feature-engineering module used by every model — no duplicated logic
- Reference-date aware, so features reflect what was knowable at prediction time
- Designed to avoid label leakage between training and evaluation splits

Predictive Models — Real Performance

4 trained models cover all 8 required predictions



Remaining predictions — completion probability, project success probability, mentor workload — are derived from these core models rather than separately trained.

Explainable AI — Not Just a Score



Every prediction is paired with a SHAP explanation — the top factors that actually drove the model's output, ranked by impact.

- Mentors see why an intern was flagged, not just that they were
- Builds trust — predictions are auditable, not a black box
- Powers the recommendation engine's reasoning directly

Example: Dropout-risk factors

Attendance rate



Task completion rate



Code review score



Communication activity



Mentor feedback score



Illustrative — actual factors vary per intern and prediction

From Prediction to Action

A rule-based recommendation engine wired directly to live model thresholds



Mentor Intervention

Triggered automatically when dropout risk crosses the tuned threshold



Easier / Advanced Tasks

Difficulty adjusted to match predicted learning speed



Learning Resources

Targeted material suggested for weak skill areas



Weekly Goals

Personalized targets generated from current performance trend

Dashboards for Every Role

Shipped as both a native web UI and a Streamlit application



Admin

- Batch-wide internship health
- High-risk students & top performers
- Department & batch analytics



Mentor

- Weak / strong student roster
- Pending reviews queue
- Severity-tiered risk alerts



Student

- Personal performance score
- AI-generated weekly plan
- Predicted internship outcome

Engineering & Observability

Fully containerized, with production-style monitoring baked in

FastAPI

PostgreSQL

Redis

Scikit-learn

XGBoost

SHAP

MLflow

Docker Compose

Prometheus

Grafana



MLflow tracks every experiment and model version. **Prometheus + Grafana** monitor live API metrics — this isn't a model trained once and forgotten; it's built to be watched in production.



Thank You

8 predictions · Explainable AI · Recommendation engine · 3 role-based dashboards · Full monitoring stack

PRESENTED BY



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Ezitech Engineering Framework — Case Study AI-005

github.com/2023-BSE-057-Saira/intern-analytics-platform